

HYDRAULIC PRESS MAINTENANCE MANUAL - PRESS-1000

1000 Ton Four-Post Hydraulic Press

SECTION 1 - MACHINE OVERVIEW

The press consists of main cylinder, four tie rods, moving platen, hydraulic power unit HPU-P1000, and PLC safety system. Cycle time 12 sec full stroke at rated tonnage.

SECTION 2 - SAFETY SYSTEMS

Light curtain LC-01, two-hand control, pressure switch proof-of-tonnage, and redundant relief valves. Test light curtain daily per shift start checklist. Never bypass safety PLC inputs.

SECTION 3 - HYDRAULIC PARAMETERS

System pressure rated: 315 bar

Tank temperature normal: 45-55 C

Pre-fill valve response: <200 ms

Decompression time: adjustable 0.5-2.0 sec to prevent shock

SECTION 4 - PREDICTIVE MONITORING

- Pressure hold test weekly: drift >5% indicates seal bypass
- Pump noise trending via ultrasonic meter monthly
- Oil particle count quarterly
- Cylinder rod coating inspection for scoring

SECTION 5 - PREVENTIVE MAINTENANCE SCHEDULE

Daily: oil level, leak check, safety test

250 hours: filter indicator check

500 hours: return filter change

2000 hours: oil sample and cooler clean

8000 hours: pump overhaul inspection

16000 hours: cylinder seal replacement (planned outage)

SECTION 6 - TROUBLESHOOTING

Slow approach speed: check pump flow, temperature, suction strainer

Hammer on decompression: adjust valve, check accumulators

Overheating: cooler fouling, relief valve passing, wrong viscosity

Tonnage loss: seal wear, pressure gauge error, PLC scaling

SECTION 7 - ALIGNMENT

Check platen parallelism annually with four corner pressure film or electronic level. Misalignment damages tooling and tie rods.

SECTION 8 - DOCUMENTATION

Log each overload trip with die number and material thickness. Pattern of trips indicates tooling or alignment issue, not random faults.

SECTION 9 - SPARE PARTS

Seal kits SK-P1000-CYL, pump cartridge PC-P1000, filters FLT-HYD-10-200. Store seals in climate-controlled area.

SECTION 10 - SHUTDOWN FOR MAINTENANCE

LOTO HPU-P1000 breakers, bleed accumulators, block cylinder with mechanical supports before working under ram.